# A Weighting Factor for Estimating the Winding Eddy Loss in Transformers for High Frequencies

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***Abstract*—In this paper, a method for estimating the winding Eddy Loss based on an improved harmonic loss factor for transformers under harmonic conditions is presented. The method considers the skin depth of penetration of the field at higher harmonic orders, leading to a greater degree an accurate estimation of the winding Eddy losses under harmonic conditions.**

***Keywords—Winding Eddy Losses, Harmonics loss factor, Harmonic conditions***

## I. INTRODUCTION

South Africa has now a generating capacity larger than a gigawatt of utility-scale solar photovoltaic (USSP) power plants, grasping the 10th position in the world for countries using solar power. The Republic of China currently is the world leader for the deployment of USSP power having 370 USSP generating plants, producing a snowballing 12GW. Other countries on top of the list include the United States of America (11GW), United Kingdom (4GW), Germany (3.6GW), India (3.3GW), Spain (1.5GW), Japan (1.4GW) Canada (1.4GW) and France (31.3GW) [1].

Being recently mentioned, this lodges South Africa in the 10th place with about 26 USSP generating plants, supplying 1.1 GW to the country's power grid. Recent reports [2], indicate that there are about 18 USSP plants under progress in South Africa. These plants are forecasted to contribute an additional 1GW capacity. The excellent solar radiation in South Africa definitely promise a bright future for USSP plants. However, the significance of harmonics and distortion in the electrical grid has heightened substantially in the last decade due to the use non-linear loads that draw current during start up and produce high frequencies. Many a lot of transformers facilitating USSP plants to supply linear loads with sinusoidal load currents have their loads progressively replaced with non-linear loads that infuse distorted harmonic currents. These harmonic current content derive increased losses and therefore bring about peculiar winding temperature rise and consequently reduce the intended service lifetime of the transformer [3], [4], [5] and [6]. As a result, the transformer requires a suitable selection of the continuous power rating (derating) or installation of a new and thrifty transformer.

The IEEE std. C57. 10-2018[7], was established on the basis to determine the capability of a transformer when during the design stage for supplying the non-linear loads if the harmonic spectrum of the USSP plants is known. Alternatively, the recommended practice can be extended to existing transformers in service. The winding Eddy loss is presumed to be progressive with the square of the harmonic order. The term harmonic loss factor was coined in efforts to describe the effective thermal effect on account of the distorted harmonic load current. As a result of skin effect phenomena, the electromagnetic flux cannot completely penetrate the winding conductor surface at high frequency. Therefore, an exponent of two utilized in the assumption that the winding Eddy losses progress with the square of the harmonic order is on the conservative side [8], [9]. The authors in [10], [11], [12], [13] and [14] have presented the harmonic factor to determine the capability of transformer under harmonic conditions. In this paper, winding Eddy loss estimation based on an improved harmonic loss factor (HLF) for transformers under harmonic conditions. A comparative study between the proposed HLF [15] and IEEE method [7] is presented.

## II. IEEE RECOMMENDED PRACTICE C57. 10-2018

### A. Review of harmonic losses

Transformers are designed such that they supply the required continuous power with maximum efficiency and low losses. The losses in transformers are categorized into the no-load ($P_{NL}$) and load loss ($P_{LL}$) as follows in eq. (1).

$$P_{TOTAL} = P_{NL} + P_{LL} \tag{1}$$

The no-load loss is caused by the induced voltage excitation in the core. This loss is reliant on the maximum flux levels essential to magnetize the core and is insignificant in terms of the current harmonics. On the other hand, the load loss can further be stated as follows in eq. (2), as an arithmetic sum of the copper loss ($P_{I2R}$), winding Eddy loss ($P_{EC}$) and the metal parts loss ($P_{MP}$).

$$P_{LL} = P_{I2R} + P_{EC} + P_{MP} \tag{2}$$

The copper loss is predominantly caused by the root mean square (rms) harmonic current and the direct current (DC) resistance. The significant amount of rms current caused by harmonics will increase the copper losses. The Eddy current losses in the transformer windings are as a result of the fluctuating electromagnetic field on the surface of the winding conductors. The skin and proximity effects are the two phenomenon's that cause additional losses in the windings [16], [17]. As a result of the excitation current, the high voltage winding produces a flux density, in which pierce through the low voltage winding and the core to induce an internal voltage that derive the circulating and Eddy currents. A transformer design philosophy is such that the internal winding, generally the low voltage winding, will have a significant amount of the Eddy current loss due to the intensity of the electromagnetic near the core casing the windings.

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According to [7], the winding Eddy losses increase with the product of the square of the harmonic current and the harmonic order. The expression for the winding Eddy losses is as follows in eq. (3).

$$P_{WE} = P_{WE_R} \times \sum_{h=1}^{h=h_{max}} \left(\frac{I_h}{I_R}\right)^2 \times h^2 \quad (3)$$

At lower harmonic order, the effect of the skin effect is minimum on the transformer winding conductors. The corresponding harmonic factor to account for the additional losses is as follows in eq. (4).

$$F_{WE} = \frac{\sum_{h=1}^{h=h_{max}} \left(\frac{I_h}{I_R}\right)^2 \times h^2}{\sum_{h=1}^{h=h_{max}} \left(\frac{I_h}{I_R}\right)^2} \quad (4)$$

All the metallic structures in the transformer active part undergo an internally induced voltage that results in the flow of Eddy currents on the surface of ferromagnetic materials. The Eddy currents generate losses that are manifested in the form of heat energy, producing supplemental temperature rise within the metallic structures and its surrounding. Other stray losses generally refer to the Eddy current losses outside from the windings. The other stray losses in the Tank Walls, Flitch plates, Core Clamps, frames, core etc. are proportional to the product of the square of the harmonic current and exponent 0.8 of the harmonic order. The expression of the other stray losses is as follows in eq. (5).

$$P_{MP} = P_{PM_R} \times \sum_{h=1}^{h=h_{max}} \left(\frac{I_h}{I_R}\right)^2 \times h^2 \quad (5)$$

The corresponding harmonic loss factor to account for the additional losses are expressed as shown in eq. (6).

$$F_{MP} = \frac{\sum_{h=1}^{h=h_{max}} \left(\frac{I_h}{I_R}\right)^2 \times h^{0.8}}{\sum_{h=1}^{h=h_{max}} \left(\frac{I_h}{I_R}\right)^2} \quad (6)$$

### *B. Maximum loading capacity*

Transformers generally have a high initial price of installation. Additionally, substituting them at the end of their service life has a huge financial and technical challenges. Accordingly, a complete inspection into the transformer secured operation leads to essential financial advantages.

$$P_{WEC(p.u)} = I^2_{(p.u)}\left(1 + F_{WEC} \times P_{EC_R(p.u)}\right) \quad (7)$$

The recommended procedure described in [7] for oil-immersed transformers is founded on the condition that the per- unit distorted harmonic current will generate some thermal heating on the winding conductors. The maximum permissible current. This effect is expressed as follows in eq. (8).

$$I_{max}(pu) = \sqrt{\frac{P_{LL-R}(p.u)}{1+F_{WEC}\times P_{WEC-R}(p.u)+F_{PM}\times P_{PM_R}(p.u)}} \quad (8)$$

This makes the assumption that the normal insulation life of the transformer will be sustained.

## III. IMPROVED HARMONIC LOSS FACTOR

At high harmonic orders, the tendency of harmonic current to flow through the outer layer of a winding conductor (skin effect) or the restriction of the electromagnetic flux to completely penetrate the winding conductors needs to be considered when estimating the winding Eddy losses. An Improved harmonic loss factor is proposed based on the equation as expressed in eq. (9) [15].

$$\Delta P_{EC} = \frac{\pi^2 f^2 T^2 B^2}{3\rho} F(\xi) \quad (9)$$

The correction function under harmonic conditions is expressed as follows in eq. (10).

$$F(\xi) = \frac{3}{\xi}\frac{\sinh\xi - \sin\zeta}{\cosh\xi - \cos\xi} \quad (10)$$

The conductor dimension related to the skin depth under rated conditions is expressed as follows in eq. (11).

$$\zeta_R = \frac{H_{cond.}}{\delta_R} \quad (11)$$

The depth of penetration ($\delta_R$) at fundamental frequency (50Hz) is expressed as follows in eq. (12)

$$\delta_R = \sqrt{\frac{\rho}{\mu\pi f}} \quad (12)$$

In rectangular copper winding conductors at fundamental frequency of 50Hz, $\delta_R = 9.7mm$. The conductor dimension related to the skin depth under harmonic conditions is expressed as follows in eq. (13).

$$\zeta_h = \zeta_R\sqrt{h} \quad (13)$$

The function ($F(\xi)$) is as demonstrated in Fig. 2 for values of $\xi$ between 0 and 10. At $\xi = 1$ The correction function $F(\xi) \approx 1$ and at low harmonic orders the winding Eddy loss is proportional to the square of the harmonic order and conductor dimensions.

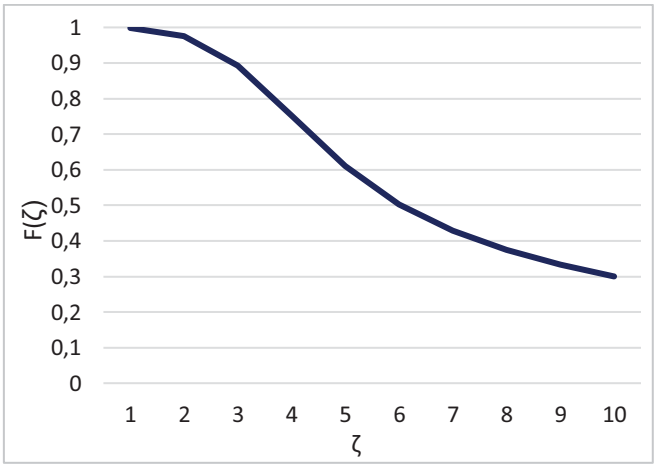



Fig. 1. The function $F(\xi)$

Standardizing of the winding Eddy loss generated by harmonic currents an improved harmonic loss factor is achieved as expressed in eq. (14).

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$$P_{MP}{}^{*} = P_{PM_R} \times \sum_{h=1}^{h=h_{max}} \left(\frac{F(\xi_h)}{F(\xi_r)}\right)\left(\frac{I_h}{I_R}\right)^2 \times h^2 \qquad (14)$$

The harmonic loss factor to account for the harmonic content that will be endured by the transformer while in service in expressed in eq. (15).

$$F_{WE}{}^{*} = \frac{\sum_{h=1}^{h=h_{max}} \left(\frac{F(\xi_h)}{F(\xi_r)}\right)\left(\frac{I_h}{I_R}\right)^2 \times h^2}{\sum_{h=1}^{h=h_{max}} \left(\frac{I_h}{I_R}\right)^2} \qquad (15)$$

The significance of the above improved harmonic factor is illustrated in fig. 3, at which the curves are demonstrated as a function of the harmonic order for various copper rectangular conductor dimensions. The curve for $h^2$ is included for comparative purposes. It is observed that for conductors with small dimensions, approach $h^2$ at low harmonic order.

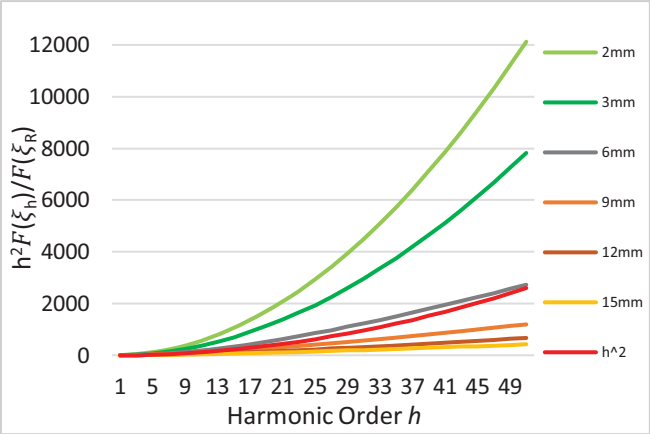



Fig. 2. The function $h^2F(\xi_h)/F(\xi_R)$ vs. harmonic order h

It can be well established from Fig. 3 that for winding conductors with small dimension, the skin effect is negligible and can be assumed to that the IEEE method estimate the winding Eddy loss with accuracy. The IEEE method compute the losses which are higher than the actual service losses for conductors with large dimension at high harmonic order as a result of the neglected limited depth of penetration. The magnetic flux impinging upon the surface of the winding conductors is minimum on account of skin effect.

## IV. Case Study: Oil Immersed Transformer

In this section, the impact of harmonic currents on oil-filled transformers serving within a harmonically contaminated environment is studied. The studied transformers supply various linear and non-linear loads during their operation period. The amount of harmonic currents seen by the transformer during service is shown in fig. 4. It is observed from the harmonic profile that the odd harmonics of the point of common coupling are significant then the even harmonics.

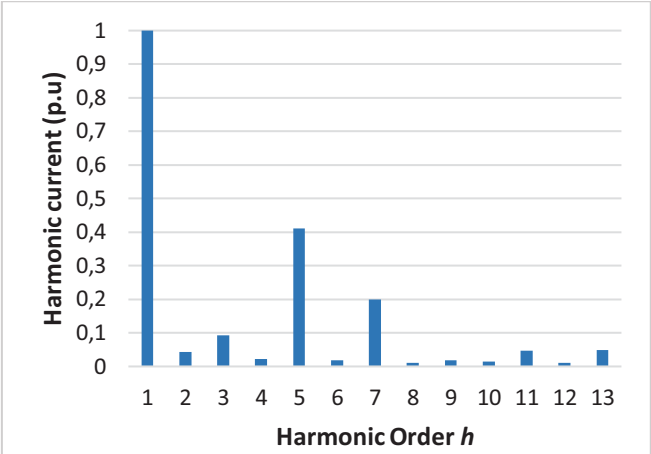



Fig. 3. Harmonic spectrum load current

The rated losses and corresponding load losses under the supplied harmonic currents are presented in TABLE I. the load losses of the oil-filled transformers are comprised of the copper loss, winding Eddy loss and Metal parts loss. The rms current in p.u under the supplied harmonic profile for transformer A and B is 1.107 and 1.224 respectively.

TABLE I. Transformer Fundamental Losses

| | Transformer A | | Transformer B | |
|---|---|---|---|---|
| **Type of Loss** | **Rated Losses(W)** | **Load Losses (W)** | **Rated Losses(W)** | **Load Losses (W)** |
| No-Load | 3,182 | 3,182 | 4,019 | 4,019 |
| $I^2R$ | 16,759 | 20,521 | 39,556 | 48,435 |
| Winding Eddy | 541 | 661.952 | 724 | 886.766 |
| Metal parts | 1,091 | 1336 | 1,550 | 1898 |
| **Total** | **21,573** | **25,701** | **45,849** | **55,239** |

A percentage breakdown of the loss distribution for transformer A is presented in Fig. 5. A large portion is observed to be the copper losses with about 78%. The winding Eddies contribute about 2.5 % on the overall losses.

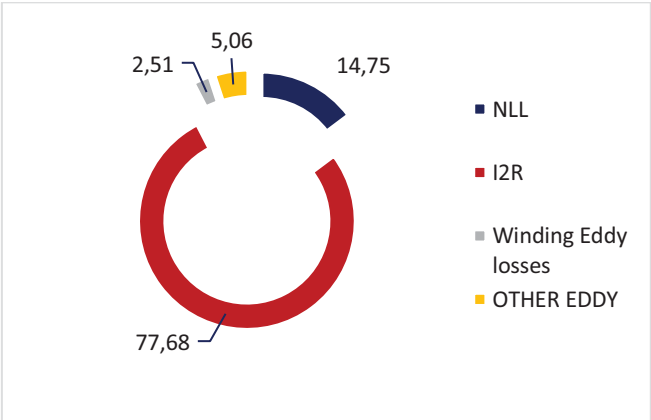



Fig. 4. Percentage Loss Distribution for transformer A

Similarly, a percentage breakdown of the loss distribution for transformer B is presented in Fig. 6. A large portion is observed to be the copper losses with about 86%. The winding Eddies contribute about 1.58 % on the overall losses.

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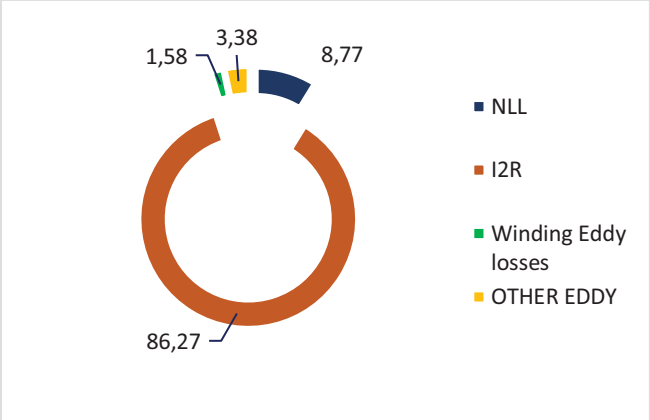



Fig. 5. Percentage Loss Distribution for transformer B

*a) Estimation of service losses*

In this section, the service losses for the transformers under study are presented taking into consideration the harmonic content their susceptible to during their service life. The winding conductor dimensions of 12.1x3.6mm and 14.2x 1.6mm are considered for transformer A and B respectively.

It is observed in TABLE II that the improved harmonic factor is reduced by 15.8% compared to the IEEE method. As a result the winding Eddy loss and total losses are reduced by 15.8% and 2.3% respectively.

TABLE II. TRANSFORMER HARMONIC LOSSES FOR TRANSFORMER A

| Type of Loss | Load Losses (W) | $F_{WE}$ | IEEE Losses | $F_{WE}^{*}$ | Corrected WEL |
|---|---|---|---|---|---|
| No-Load | 3,182 | - | 3,182 | - | 3,182 |
| $I^2R$ | 20,521 | - | 20,521 | - | 20,521 |
| Winding Eddy | 661.952 | 6.52 | 4,314 | 5.490 | 3,634 |
| Metal parts | 1336 | 1.52 | 2,033 | - | 2,033 |
| **Total** | **25,701** | | **30,050** | | **29,370** |

As observed in TABLE III, the harmonic factors are 5.78 and 4.306 for the IEEE and Improved method respectively. As a result the winding Eddy losses are reduced by 34% as a result of the improved harmonic factor and an overall reduction of 3.2% on the total losses.

TABLE III. TRANSFORMER HARMONIC LOSSES FOR TRANSFORMER B

| Type of Loss | Load Losses (W) | $F_{WE}$ | IEEE Losses | $F_{WE}^{*}$ | Corrected WEL |
|---|---|---|---|---|---|
| No-Load | 4,019 | - | 4,019 | - | 4,019 |
| $I^2R$ | 48,435 | - | 48,435 | - | 48,435 |
| Winding Eddy | 886.766 | 6.52 | 5,780 | 4.306 | 3,819 |
| Metal parts | 1898 | 1.52 | 2,887 | - | 2,887 |
| **Total** | **55,239** | | **61,121** | | **59,160** |

The results in this section, indicate that the effect of skin effect at high harmonic order plays a pivotal role in the estimation of the winding Eddy losses. A significant reduction of the losses is observed due to the limitation of the magnetic flux to completely penetrate the winding conductors at high harmonic order. This effect is neglected on the IEEE std. C57. 10-2018, assuming an increase in the winding Eddy loss with the square of the harmonic order.

*b) Transformer capacity*

In this section, to account for the most undesirable thermal operating conditions of a transformer, the winding Eddy loss can be computed using eq. (7) and eq.(8) as shown in TABLE IV.

TABLE IV. CAPACITY FOR TRANSFORMER A

| Method | Pec (p.u) | Imax(p.u) | Imax |
|---|---|---|---|
| IEEE | 1.482 | 0.916 | 601 |
| Improved | 1.224 | 0.927 | 608 |

The maximum permissible current (MPC) as calculated in TABLE IV determines the maximum current that can flow through the winding conductors without a significant degradation of the insulation. The computation of the MPC allows the utility owners to protect the transformer from reduction of the intended service lifetime by regulating its MPC. From TABLE IV, the IEEE and improved method have a MPC percentage error of about 8.45% and 7.27% when compared to the rated current. Similarly both methods have a percentage error of 1.274%, while the winding Eddy loss has a percentage error of 17.37%. The capability of transformer B are shown in TABLE V.

TABLE V. CAPACITY FOR TRANSFORMER B

| Method | Pec (p.u) | Imax(p.u) | Imax |
|---|---|---|---|
| IEEE | 1.371 | 0.947 | 621 |
| Improved | 1.224 | 0.963789907 | 632 |

For transformer B, based on the IEEE and improved harmonic factor method, there is a percentage error of about 5.29% and 3.62% on the maximum current when compared against the rated current. On the other hand, the both methods have a percentage error of about 1.73%. Lastly, the winding Eddy losses Pec (p.u) have a percentage error of about 10.66%.

## V. CONCLUSION

The explanation given by the IEEE std. C57. 10-2018 that provides a reason for observed increase in winding Eddy losses under harmonic conditions as proportional to the square of the harmonic order does not hold for all conductor dimensions. This method is reasonably accurate for conductors with small dimensions. In the case of winding conductors with large dimensions, such assumption leads to conservative results. Due to the neglected depth of penetration, the predicted losses will be higher than the actual service losses. In reality the magnetic flux will not completely penetrate the conductors at high harmonic orders due to the skin effect.

An improved harmonic loss factor which regards the depth of penetration of the magnetic field upon the surface of the winding conductors can be applied to estimate more accurately the capability of transformers under harmonic conditions.

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